

Technical Appendix: Sampling-Boundary Distortion in Sharpe Ratio: Evidence from Financial Panels

1 Purpose

This appendix provides derivations, calibration checks, robustness tables, and reproducibility details omitted from the main manuscript for brevity. It supports the proof, calibration, and robustness record; it is not an independent empirical claim.

2 Public-Panel Momentum Implementation

The public French panel implementation uses a skipped-momentum signal as the running row-level illustration. The sampling-boundary diagnostic does not depend on this particular signal, but the formulas below define the exact public-panel construction used in the reported tables. For lookback L and skip q , the skipped compounded signal is

$$s_{t,i}^{(L,q)} = \prod_{j=t-L-q+1}^{t-q} (1 + r_{j,i}) - 1. \quad (1)$$

With $q \geq 1$, the signal excludes the outcome return and is executable at date t when all signals in the available universe A_t are known before the trading decision. The signed one-period row return is

$$y_{t+1,i}^{(L,q)} = \text{sign}(s_{t,i}^{(L,q)}) r_{t+1,i}, \quad (2)$$

with $\text{sign}(0) = 0$; zero-signal and missing-signal rows receive zero economic weight. The confidence score $c_{t,i}^{(L,q)}$ is the within-date percentile rank of $|s_{t,i}^{(L,q)}|$, and the selected set is

$$\mathcal{S}_t(L, q, \tau) = \{i : c_{t,i}^{(L,q)} \geq \tau\}.$$

The percentile score is a baseline screening rule, not an optimal weighting theorem. The replication files record tie handling, one-name dates, selected-count granularity, and source-specific frequency and annualization metadata.

3 Mean-Return Design-Effect Proof

For a selected row model $y_{t,i} = \mu + \lambda_t + \epsilon_{t,i}$ with constant selected count $\bar{n}$, same-date intraclass correlation ρ , and independent dates, the row-pooled mean equals the average of date means:

$$\hat{\mu} = \frac{1}{T\bar{n}} \sum_{t=1}^T \sum_{i=1}^{\bar{n}} y_{t,i} = \frac{1}{T} \sum_{t=1}^T \bar{y}_t.$$

The row-naive variance report is $\sigma^2/(T\bar{n})$. The true one-date variance is

$$\mathbb{V}\text{ar}(\bar{y}_t) = \frac{1}{\bar{n}^2} \{ \bar{n}\sigma^2 + \bar{n}(\bar{n} - 1)\rho\sigma^2 \} = \frac{\sigma^2}{\bar{n}} \{ 1 + (\bar{n} - 1)\rho \}.$$

Thus

$$\frac{se_{date,true}}{se_{row,naive}} = \sqrt{1 + (\bar{n} - 1)\rho}.$$

This is the Moulton-like design effect used as the mean-return special case in the main text. It is a closed-form special case, not the full Sharpe-ratio distortion result.

4 Sharpe UVIF Derivation

These calculations supply the algebra behind the Sharpe UVIF decomposition in the main manuscript. Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. The annualized Sharpe ratio is the smooth functional

$$f(\mu, m_2) = \sqrt{A_f} \frac{\mu}{(m_2 - \mu^2)^{1/2}}, \quad d_R = \nabla f(\mu_R, m_{2,R}).$$

The annualization factor A_f is source-specific: the public configuration uses 252 for daily sources and 12 for monthly sources. On sets where $v_R = m_{2,R} - \mu_R^2 > 0$, the first-order expansion is

$$\widehat{\text{SR}}_R - \text{SR}_R = d'_R(\hat{\theta}_R - \theta_R) + o_p(T_R^{-1/2}), \quad \theta_R = \mathbb{E}[G_t].$$

With long-run covariance $\Omega_R = \Gamma_0^R + \sum_{k \geq 1} (\Gamma_k^R + \Gamma_k^{R'})$,

$$\mathbb{V}\text{ar}_{LR}(\widehat{\text{SR}}_R) = T_R^{-1} d'_R \Omega_R d_R + o(T_R^{-1}).$$

The row-pooled IID calculation applies the same delta method to $\bar{n}T_R$ row observations $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$:

$$\mathbb{V}\text{ar}_{IID,row}(\widehat{\text{SR}}_{row}) = (\bar{n}T_R)^{-1} d'_Y \Psi_Y d_Y + o((\bar{n}T_R)^{-1}).$$

Taking the ratio gives the Sharpe UVIF:

$$\text{UVIF}_{\text{SR}} = \frac{\bar{n} d'_R \Omega_R d_R}{d'_Y \Psi_Y d_Y} = \frac{\bar{n} d'_R \Gamma_0^R d_R}{d'_Y \Psi_Y d_Y} \times \frac{d'_R \Omega_R d_R}{d'_R \Gamma_0^R d_R}.$$

The first term is the cross-sectional component and the second term is the serial long-run factor. Componentwise,

$$d'_R \Omega_R d_R = \frac{A_f}{v_R^3} \left[m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right].$$

The Ω_{22} term contains long-run covariances of squared returns and therefore brings fourth moments into Sharpe-ratio variance.

5 Optional Order-Size Sensitivity

The public sorted-portfolio examples in the main manuscript do not contain average daily volume or order-size information, so they do not support an execution-scale claim. For traded-security panels where those inputs are available, a square-root impact diagnostic can be added as an optional sensitivity extension.

Proposition 1 (Square-root order-size sensitivity bound). *Suppose a traded panel has row-pooled gross mean $\bar{r}_{\text{gross}}$, row-pooled return standard deviation σ_{row} , strictly positive average turnover $\widehat{\text{TO}}$, T trading dates, square-root impact cost $\lambda \widehat{\text{TO}} \sqrt{A/V}$, and a dependence standard-error inflation factor $\mathcal{M} \geq 1$ relative to row-pooled inference. For a one-sided level- α sensitivity margin, any statistically screened asset scale A^* must satisfy*

$$A^* \leq V \left(\frac{\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}}{\lambda \widehat{\text{TO}}} \right)^2 \quad (3)$$

when the numerator is positive; otherwise no positive scale is screened by the margin.

For a square-root impact diagnostic, the dependence-adjusted one-sided lower confidence margin for gross mean return is

$$\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T},$$

where σ_{row} is the standard deviation of row-pooled returns before date aggregation or UVIF adjustment. Square-root impact subtracts $\lambda \widehat{\text{TO}} \sqrt{A/V}$. Requiring the net lower margin to be nonnegative gives

$$\lambda \widehat{\text{TO}} \sqrt{A/V} \leq \bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}.$$

Rearranging yields the sensitivity bound in Proposition 1. If the volatility input is already the date-level portfolio volatility, the date boundary has already embedded the cross-sectional UVIF component, so the cross-sectional part of $\mathcal{M}$ must not be multiplied in again.

6 Higher-Moment UVIF and ERIR Details

For $\theta^{(4)} = (\mu, m_2, m_3, m_4)'$, define skewness and kurtosis from the central moments implied by $\theta^{(4)}$. Any adjusted Sharpe functional $h(\theta^{(4)})$ has a delta-method variance $\nabla h' \Omega^{(4)} \nabla h$, where $\Omega^{(4)}$ is the long-run covariance matrix of $(R_t, R_t^2, R_t^3, R_t^4)'$. The higher-moment UVIF is the corresponding ratio of date-level long-run variance to row-pooled IID variance. It is a diagnostic extension, not an additional empirical rejection rule.

The Effective Row Information Retention (ERIR) diagnostic is a design-effect retention calculation, not a hypothesis test or a formal information claim. Under the same equicorrelated approximation used for the mean-return design effect,

$$DE_{\mu, xs} = 1 + (\bar{n} - 1)\rho, \quad \text{ERIR} = \{1 + (\bar{n} - 1)\rho\}^{-1}.$$

Thus an ERIR of 0.111 means that roughly $1 - 0.111 = 0.889$, or 89 percent, of nominal row-level information is redundant under this diagnostic approximation. Confirmatory inference remains based on date-level HAC-delta and dependent-bootstrap procedures. When the descriptive equicorrelation estimate is non-positive, the unclipped ERIR can exceed one and the corresponding UVIF can fall below one; the main manuscript displays a conservative one-sided UVIF floor at one and reports the unclipped diagnostics separately.

7 Synthetic Positive Control

This section gives the appendix version of the known-edge calibration check. The synthetic panel is not empirical evidence for a public strategy; it verifies that the pre-specified reporting rule can accept a declared positive signal when the data-generating process contains one. The generated panel has $T = 5000$ dates and $N = 100$ names. Signed returns include daily alpha 0.12/252, idiosyncratic volatility 0.01, a common loading of 0.50, AR(1) common-factor coefficient 0.10, signal-stay probability 0.98, and seed 777. The displayed run is one representative known-edge calibration with 5000 bootstrap draws and 10000 same-date permutations; costs use the same 5 bps turnover-stress convention as the public tables. The replication screen below uses fewer resampling draws and is

reported only as a power sanity check. The data-generating process is

$$f_t = 0.10f_{t-1} + u_t, \quad \epsilon_{t,i} \sim N(0, 0.01^2),$$

$$z_{t,i} \in \{-1, 1\}, \quad \Pr(z_{t,i} = z_{t-1,i}) = 0.98,$$

$$r_{t+1,i} = z_{t,i}(0.12/252) + 0.50f_t + \epsilon_{t,i}, \quad s_{t,i} = z_{t,i}.$$

The calibration then applies the same signed-row construction, date aggregation, HAC-delta inference, Romano-Wolf menu adjustment, same-date permutation, and 5 bps cost stress used in the public tables.

Table 1. Synthetic positive-control evaluation.

Panel	$\bar{\rho}$	SE infl.	SR	HAC p^+	RW p	Perm. p^+
Synthetic control	0.202	4.58	1.744	7.67e-13	2.00e-04	1.00e-04

Table 2. Positive-control replication screen.

Reps	Boot	Perms	Block pass	HAC pass	RW pass	Perm. pass	Cost pass	All pass
100	499	199	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

Note: The replication screen uses fewer resampling draws than the detailed calibration row and is reported only as a power sanity check.

8 Full Empirical Tables

These tables are supporting diagnostics rather than primary evidence; the main text reports the compressed interpretation. The inferential tables in this appendix report only public sources that loaded and produced reproducible outputs. Failed source-access attempts remain in the generated provenance files and run metadata, but are not treated as empirical evidence. Panel candidates and pre-aggregated factor benchmarks are reported in separate groups because only the former provide row-level signals for same-date permutation and sampling-boundary diagnostics. For each empirical table group, the check is whether the result survives moving from rows to date returns, whether nearby HAC and dependent-bootstrap variants agree, and whether placebo or cost diagnostics overturn the simple Sharpe interpretation. Large disagreement across these tables is a robustness warning, not a new discovery channel.

This supplement reports loaded sources in separated groups. The loader provenance is retained in the generated metadata; it is not a comparable pass/fail table. The tables are supporting diagnostics rather than primary evidence; the main text reports the compressed interpretation.

Table 3. Source frequency and annualization metadata.

Source	type	period	A_f	dates
AQR Betting Against Beta, equity factors	single_series_factor	daily	252.000	25281
AQR HML Devil	single_series_factor	daily	252.000	26590
AQR Quality Minus Junk	single_series_factor	daily	252.000	17681
AQR Value and Momentum Everywhere	single_series_factor	monthly	12.000	651
French momentum deciles, rank-threshold panel	panel	daily	252.000	26131
French WML decile benchmark	panel	daily	252.000	26131
French 25 Size/BM, skipped-momentum panel	panel	daily	252.000	3477

Table 4. Panel candidates with row-level signal-return structure.

Panel candidate	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Full rule 5%	Applicable checks 5%
French momentum deciles, rank-threshold panel	0.447	7.54e-05	2.00e-04	2.00e-04	N/A	0.447	no	yes
French WML decile benchmark	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes	yes
French 25 Size/BM, skipped-momentum panel	0.058	0.412	0.422	0.480	0.943	-0.050	no	no

Table 5. Single-series factor benchmarks subject to time-series inference only.

Single-series benchmark	SR	HAC p+	Boot p+	RW p	Scope
AQR Betting Against Beta, equity factors	0.741	3.08e-13	2.00e-04	2.00e-04	time-series only
AQR Quality Minus Junk	0.536	2.26e-05	2.00e-04	2.00e-04	time-series only
AQR HML Devil	0.336	0.004	0.002	0.002	time-series only
AQR Value and Momentum Everywhere	0.572	1.62e-04	2.00e-04	2.00e-04	time-series only

Note: Time-series only means not eligible for same-date permutation, row-retention diagnostics, or the full panel decision rule.

Table 6. Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
French WML decile benchmark	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

Table 7. Comparison with common row and date-level inference choices for panel candidates.

Panel candidate	Row-naive p+	Date-IID p+	Date-HAC p+	Block-bootstrap p+	Sharpe UVIF
French momentum deciles, rank-threshold panel	1.70e-04	2.00e-04	7.54e-05	2.00e-04	1.000
French WML decile benchmark	4.85e-04	2.00e-04	5.53e-06	2.00e-04	1.000
French 25 Size/BM, skipped-momentum panel	0.253	0.417	0.412	0.422	9.002

Note: The row-naive column treats selected rows as independent. Date-IID, date-HAC, and block-bootstrap columns operate on the date-level portfolio return series.

Table 8. Selected-count and dependence diagnostics for panel candidates.

Candidate	status	thr	rows	dates	avg names	rho	SE infl.
French momentum deciles, rank-threshold panel	Available	0.700	104524	26131	4.000	-0.138	0.766
French WML decile benchmark	Available	0.900	52262	26131	2.000	-0.583	0.646
French 25 Size/BM, skipped-momentum panel	Available	0.500	45201	3477	13.000	0.667	3.000

Table 9. Row-naive and date-level p-value diagnostics for panel candidates.

Candidate	thr	row t	HAC z	row p+	HAC p+	RW p
French momentum deciles, rank-threshold panel	0.700	3.584	3.790	1.70e-04	7.54e-05	2.00e-04
French WML decile benchmark	0.900	3.299	4.395	4.85e-04	5.53e-06	2.00e-04
French 25 Size/BM, skipped-momentum panel	0.500	0.664	0.222	0.253	0.412	0.480

Table 10. Unclipped and one-sided floored UVIF diagnostics for panel candidates.

Candidate	rho	SE infl.	UVIF unclipped	UVIF displayed	ERIR unclipped	ERIR displayed
French momentum deciles, rank-threshold panel	-0.138	0.766	0.587	1.000	1.704	1.000
French WML decile benchmark	-0.583	0.646	0.417	1.000	2.399	1.000
French 25 Size/BM, skipped-momentum panel	0.667	3.000	9.002	9.002	0.111	0.111

Note: Displayed main-text UVIF values are floored at one when the equicorrelation diagnostic is non-inflationary; unclipped values remain descriptive diagnostics and are not rejection rules.

Table 11. Sampling-boundary summary for panel candidates.

Candidate	Row p+	UVIF	Max p+	Net SR	Status
French momentum deciles, rank-threshold panel	1.70e-04	1.000	2.00e-04	0.447	Permutation uninformative; remaining criteria pass
French WML decile benchmark	4.85e-04	1.000	2.00e-04	0.497	Passes full evaluation
French 25 Size/BM, skipped-momentum panel	0.253	9.002	0.943	-0.050	Fails statistical and cost checks
<i>Note:</i> Displayed UVIF is a one-sided inflation summary floored at one. Degenerate same-date permutation diagnostics are reported as uninformative rather than converted into failures.					

Table 12. Horizon-effect sensitivity for the dynamic Size/BM momentum panel.

L	thr	dates	avg names	gross SR	net SR 5bps	turnover	UVIF SR	serial UVIF	HAC p+
1	0.500	3497	12.913	0.073	-0.394	0.733	11.498	0.891	0.386
5	0.500	3493	13.000	-0.312	-0.528	0.339	12.880	0.990	0.878
21	0.500	3477	13.000	0.058	-0.050	0.169	12.121	0.933	0.412

Table 13. Primary-threshold dependence-aware Sharpe inference.

Candidate	thr	Method	SR	SE	CI low	CI high	p+	n
AQR Betting Against Beta, equity factors	0.000	block	0.741	0.106	0.533	0.951	2.00e-04	25281
AQR Betting Against Beta, equity factors	0.000	HAC	0.741	0.103	0.540	0.943	3.08e-13	25281
AQR Betting Against Beta, equity factors	0.000	prewhite	0.741	0.104	0.538	0.945	4.38e-13	25281
AQR HML Devil	0.000	block	0.336	0.114	0.105	0.552	0.002	26590
AQR HML Devil	0.000	HAC	0.336	0.126	0.089	0.583	0.004	26590
AQR HML Devil	0.000	prewhite	0.336	0.111	0.119	0.553	0.001	26590
AQR Quality Minus Junk	0.000	block	0.536	0.133	0.276	0.795	2.00e-04	17681
AQR Quality Minus Junk	0.000	HAC	0.536	0.131	0.279	0.794	2.26e-05	17681
AQR Quality Minus Junk	0.000	prewhite	0.536	0.128	0.285	0.787	1.40e-05	17681
AQR Value and Momentum Everywhere	0.000	block	0.572	0.156	0.268	0.882	2.00e-04	651
AQR Value and Momentum Everywhere	0.000	HAC	0.573	0.159	0.261	0.885	1.62e-04	651
AQR Value and Momentum Everywhere	0.000	prewhite	0.573	0.158	0.263	0.882	1.42e-04	651
French momentum deciles, rank-threshold panel	0.700	block	0.447	0.114	0.222	0.667	2.00e-04	26131
French momentum deciles, rank-threshold panel	0.700	HAC	0.447	0.118	0.216	0.679	7.54e-05	26131
French momentum deciles, rank-threshold panel	0.700	prewhite	0.447	0.114	0.224	0.670	4.18e-05	26131
French WML decile benchmark	0.900	block	0.497	0.111	0.275	0.708	2.00e-04	26131
French WML decile benchmark	0.900	HAC	0.497	0.113	0.275	0.718	5.53e-06	26131
French WML decile benchmark	0.900	prewhite	0.497	0.111	0.279	0.715	3.99e-06	26131
French 25 Size/BM, skipped-momentum panel	0.500	block	0.058	0.261	-0.483	0.540	0.422	3477
French 25 Size/BM, skipped-momentum panel	0.500	HAC	0.058	0.260	-0.452	0.567	0.412	3477
French 25 Size/BM, skipped-momentum panel	0.500	prewhite	0.058	0.255	-0.442	0.557	0.410	3477

Table 14. Same-date signal-permutation diagnostics for panel candidates.

Candidate	thr	obs SR	null mean	null sd	p+	status	perms
French momentum deciles, rank-threshold panel	0.000	0.395	0.410	0.039	0.646	Available	10000
French momentum deciles, rank-threshold panel	0.300	0.419	0.434	0.024	0.733	Available	10000
French momentum deciles, rank-threshold panel	0.500	0.419	0.434	0.024	0.733	Available	10000
French momentum deciles, rank-threshold panel	0.700	0.447	0.447	1.11e-16	N/A	Permutation null degenerate	10000
French momentum deciles, rank-threshold panel	0.900	0.497	0.404	0.042	0.015	Available	10000
French WML decile benchmark	0.900	0.497	-0.001	0.098	1.00e-04	Available	10000
French 25 Size/BM, skipped-momentum panel	0.000	0.040	0.080	0.017	0.991	Available	10000
French 25 Size/BM, skipped-momentum panel	0.300	0.034	0.097	0.023	0.997	Available	10000
French 25 Size/BM, skipped-momentum panel	0.500	0.058	0.104	0.029	0.943	Available	10000
French 25 Size/BM, skipped-momentum panel	0.700	0.087	0.137	0.041	0.892	Available	10000
French 25 Size/BM, skipped-momentum panel	0.900	0.054	0.119	0.069	0.832	Available	10000

Note: A near-zero permutation-null standard deviation is reported as N/A because the placebo is uninformative for that threshold, not calibrated evidence for or against predictability.

Table 15. HAC bandwidth, prewhitening, and fixed-b robustness diagnostics.

Candidate	Diagnostic	SR	SE	p+
AQR Betting Against Beta, equity factors	HAC K=auto	0.741	0.103	3.08e-13
AQR Betting Against Beta, equity factors	HAC K=21	0.741	0.119	2.04e-10
AQR Betting Against Beta, equity factors	HAC K=63	0.741	0.130	6.06e-09
AQR Betting Against Beta, equity factors	prewhite	0.741	0.104	4.38e-13
AQR Betting Against Beta, equity factors	fixed-b 0.1	0.741	0.166	0.002
AQR HML Devil	HAC K=auto	0.336	0.126	0.004
AQR HML Devil	HAC K=21	0.336	0.128	0.004
AQR HML Devil	HAC K=63	0.336	0.137	0.007
AQR HML Devil	prewhite	0.336	0.111	0.001
AQR HML Devil	fixed-b 0.1	0.336	0.110	0.004
AQR Quality Minus Junk	HAC K=auto	0.536	0.131	2.26e-05
AQR Quality Minus Junk	HAC K=21	0.536	0.140	6.33e-05
AQR Quality Minus Junk	HAC K=63	0.536	0.147	1.32e-04
AQR Quality Minus Junk	prewhite	0.536	0.128	1.40e-05
AQR Quality Minus Junk	fixed-b 0.1	0.536	0.151	0.002
AQR Value and Momentum Everywhere	HAC K=auto	0.573	0.159	1.62e-04
AQR Value and Momentum Everywhere	HAC K=21	0.573	0.155	1.14e-04
AQR Value and Momentum Everywhere	HAC K=63	0.573	0.144	3.41e-05
AQR Value and Momentum Everywhere	prewhite	0.573	0.158	1.42e-04
AQR Value and Momentum Everywhere	fixed-b 0.1	0.573	0.143	0.004
French momentum deciles, rank-threshold panel	HAC K=auto	0.447	0.118	7.54e-05
French momentum deciles, rank-threshold panel	HAC K=21	0.447	0.120	9.94e-05
French momentum deciles, rank-threshold panel	HAC K=63	0.447	0.122	1.21e-04
French momentum deciles, rank-threshold panel	prewhite	0.447	0.114	4.18e-05
French momentum deciles, rank-threshold panel	fixed-b 0.1	0.447	0.143	0.004
French WML decile benchmark	HAC K=auto	0.497	0.113	5.53e-06
French WML decile benchmark	HAC K=21	0.497	0.116	9.62e-06
French WML decile benchmark	HAC K=63	0.497	0.117	1.01e-05
French WML decile benchmark	prewhite	0.497	0.111	3.99e-06
French WML decile benchmark	fixed-b 0.1	0.497	0.147	0.002
French 25 Size/BM, skipped-momentum panel	HAC K=auto	0.058	0.260	0.412
French 25 Size/BM, skipped-momentum panel	HAC K=21	0.058	0.249	0.409
French 25 Size/BM, skipped-momentum panel	HAC K=63	0.058	0.231	0.401
French 25 Size/BM, skipped-momentum panel	prewhite	0.058	0.255	0.410
French 25 Size/BM, skipped-momentum panel	fixed-b 0.1	0.058	0.172	0.360

Table 16. Decision-rule sensitivity by alpha threshold.

alpha	candidates	full passes	applicable passes
0.010	8	1	6
0.050	8	1	6

Table 17. Holdout and subperiod Sharpe diagnostics.

Candidate	window	SR	n
AQR Betting Against Beta, equity factors	train_70pct	0.887	17696
AQR Betting Against Beta, equity factors	holdout_30pct	0.531	7585
AQR Betting Against Beta, equity factors	quarter_min	0.280	6320
AQR Betting Against Beta, equity factors	quarter_max	1.270	6320
AQR HML Devil	train_70pct	0.465	18613
AQR HML Devil	holdout_30pct	0.120	7977
AQR HML Devil	quarter_min	0.273	6647
AQR HML Devil	quarter_max	0.445	6647
AQR Quality Minus Junk	train_70pct	0.794	12376
AQR Quality Minus Junk	holdout_30pct	0.305	5305
AQR Quality Minus Junk	quarter_min	0.174	4421
AQR Quality Minus Junk	quarter_max	1.083	4420
AQR Value and Momentum Everywhere	train_70pct	0.626	455
AQR Value and Momentum Everywhere	holdout_30pct	0.431	196
AQR Value and Momentum Everywhere	quarter_min	0.372	163
AQR Value and Momentum Everywhere	quarter_max	0.941	163
French momentum deciles, rank-threshold panel	train_70pct	0.615	18292
French momentum deciles, rank-threshold panel	holdout_30pct	0.229	7839
French momentum deciles, rank-threshold panel	quarter_min	0.171	6532
French momentum deciles, rank-threshold panel	quarter_max	1.280	6533
French WML decile benchmark	train_70pct	0.641	18292
French WML decile benchmark	holdout_30pct	0.310	7839
French WML decile benchmark	quarter_min	0.224	6533
French WML decile benchmark	quarter_max	1.370	6533
French 25 Size/BM, skipped-momentum panel	train_70pct	0.088	2428
French 25 Size/BM, skipped-momentum panel	holdout_30pct	-0.007	1049
French 25 Size/BM, skipped-momentum panel	quarter_min	-0.581	853
French 25 Size/BM, skipped-momentum panel	quarter_max	0.210	875

Table 18. Turnover-scaled cost sensitivity and break-even cost.

Candidate	cost bps	turnover	gross SR	net SR	break-even bps
AQR Betting Against Beta, equity factors	0	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	1	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	5	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	10	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	25	0.000	0.741	0.741	
AQR HML Devil	0	0.000	0.336	0.336	
AQR HML Devil	1	0.000	0.336	0.336	
AQR HML Devil	5	0.000	0.336	0.336	
AQR HML Devil	10	0.000	0.336	0.336	
AQR HML Devil	25	0.000	0.336	0.336	
AQR Quality Minus Junk	0	0.000	0.536	0.536	
AQR Quality Minus Junk	1	0.000	0.536	0.536	
AQR Quality Minus Junk	5	0.000	0.536	0.536	
AQR Quality Minus Junk	10	0.000	0.536	0.536	
AQR Quality Minus Junk	25	0.000	0.536	0.536	
AQR Value and Momentum Everywhere	0	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	1	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	5	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	10	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	25	0.000	0.572	0.572	
French momentum deciles, rank-threshold panel	0	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	1	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	5	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	10	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	25	1.91e-05	0.447	0.447	85893.000
French WML decile benchmark	0	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	1	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	5	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	10	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	25	1.91e-05	0.497	0.497	122232.000
French 25 Size/BM, skipped-momentum panel	0	0.169	0.058	0.058	2.683
French 25 Size/BM, skipped-momentum panel	1	0.169	0.058	0.036	2.683
French 25 Size/BM, skipped-momentum panel	5	0.169	0.058	-0.050	2.683
French 25 Size/BM, skipped-momentum panel	10	0.169	0.058	-0.157	2.683
French 25 Size/BM, skipped-momentum panel	25	0.169	0.058	-0.480	2.683

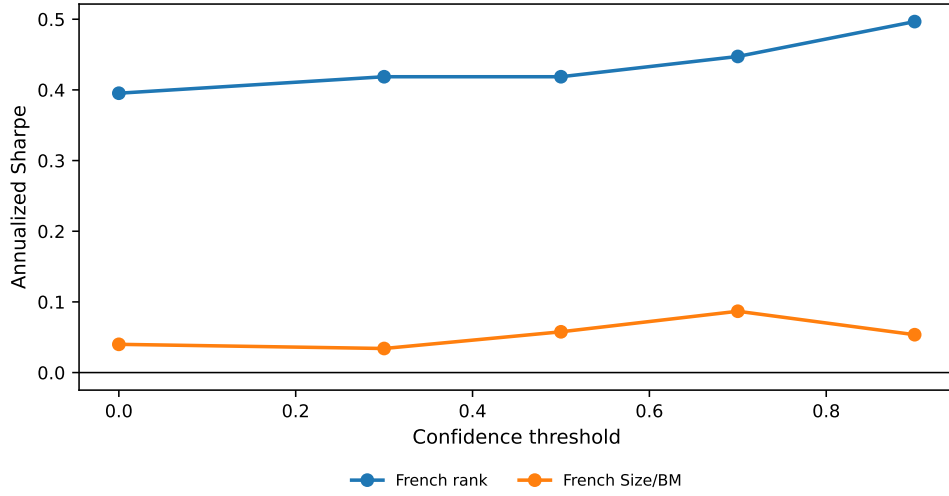


Figure 1. Threshold-profile diagnostics for public candidate panels; each line shows how annualized Sharpe changes when the pre-specified confidence threshold is varied.

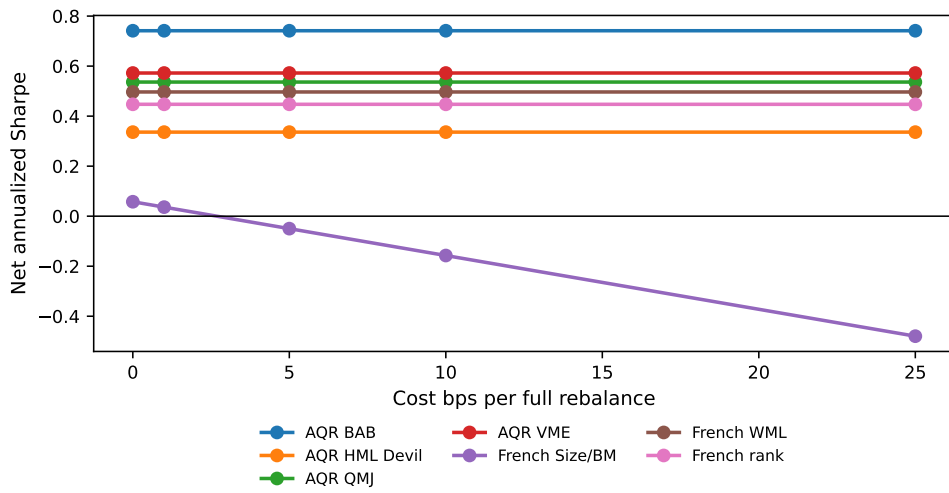


Figure 2. Turnover-scaled cost sensitivity at primary candidate thresholds; positive values indicate that the candidate remains above zero after the displayed linear cost stress.

9 Full Monte Carlo Tables

The simulation tables target the date-level Sharpe claim. Row-naïve coverage is therefore reported against that date-level target, not against a separate row-distribution target. The design-sweep table is a stress diagnostic over same-date correlation and selected-count pressure under a null design with no true edge. The main concern in this table group is size distortion: a method that looks powerful because it overrejects under the null is not acceptable for confirmatory evaluation.

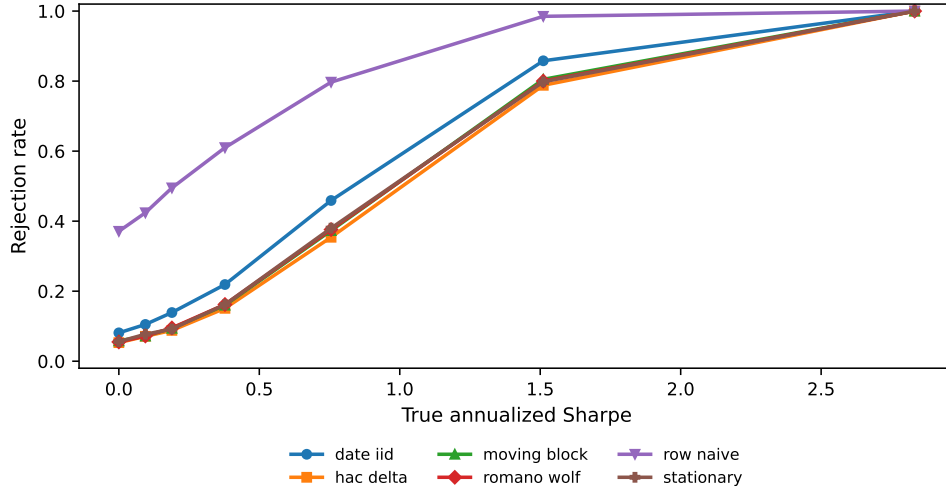


Figure 3. Monte Carlo rejection rates by true annualized Sharpe; the figure separates genuine power from methods that already overreject under null designs with no true edge.

Table 19. Monte Carlo design contract.

Design	T	N	Selected count	XS dep.	Serial dep.	Heterosk.	Reps	Resampling	Seed
IID null	1000	50	all rows	$\rho=0$	$AR(1)=0$	none	1000	5000 boot	rep index
Dependent null	1000	50	all rows	$\rho=0.2$	$AR(1)=0.2$	none	1000	5000 boot	rep index
GARCH null	1000	50	all rows	$\rho=0.2$	$AR(1)=0.2$	GARCH(0.05,0.90)	1000	5000 boot	rep index
Positive control	5000	100	signal-selected rows	common loading=0.50	$AR(1)=0.10$	none	100	499 boot	777

Note: All null designs have true Sharpe equal to zero. Positive-control rows use a declared nonzero signal and are included to show that the reporting rule is not mechanically anti-discovery.

10 Robustness and Configuration Notes

The public configuration is fixed in `code/campaign_registry.json`. It records loaders, public source metadata, threshold menus, primary thresholds, and lookback/skip parameters. Changing a source, lookback, threshold, cost assumption, or eligibility rule after seeing results creates a new researcher menu and must be reported as exploratory. Rendered tables in the submitted package use the following SHA256 identifiers:

Table 20. Monte Carlo and resampling settings.

Setting	Value
Monte Carlo replications	1000
Bootstrap draws	5000
Permutation draws	10000
Main null panel sizes	T=1000, N=50
Cross-sectional correlation grid	0.000, 0.200, 0.250, 0.300, 0.350, 0.400, 0.500
AR(1) grid	0.000, 0.150, 0.200, 0.300, 0.500, 0.700
GARCH parameter pairs	(0.050, 0.900); (0.100, 0.850); (0.150, 0.800)
HAC bandwidth rule	automatic Bartlett, with fixed-lag sensitivity checks
Moving-block length	ceil(T to the 1/5 times max(D,1)), capped at floor(T/3)
Stationary mean block	same length as moving-block rule
Dependence factor D	(2 rho1/(1-rho1)) to the 2/3 for positive AR(1) rho1; otherwise 1
Seed policy	seed-indexed replications from campaign runner

Table 21. Simulation DGP parameter grid.

DGP	Name	T	N	mu	sigma	AR1	rho	GARCH a	GARCH b	K	regime
01_iid	IID baseline	1000	50	4.00e-04	0.015	0.000	0.000	0.000	0.000	1	no
02_ser_mild	Serial mild	1000	50	4.00e-04	0.015	0.200	0.000	0.000	0.000	1	no
03_ser_strong	Serial strong	1000	50	4.00e-04	0.015	0.500	0.000	0.000	0.000	1	no
04_xs_mild	Cross mild	1000	50	4.00e-04	0.015	0.000	0.200	0.000	0.000	1	no
05_xs_strong	Cross strong	1000	50	4.00e-04	0.015	0.000	0.500	0.000	0.000	1	no
06_both_mod	Both moderate	1000	50	4.00e-04	0.015	0.200	0.200	0.000	0.000	1	no
07_both_strong	Both strong	1000	50	4.00e-04	0.015	0.500	0.500	0.000	0.000	1	no
08_realistic	Realistic	2500	100	4.00e-04	0.015	0.150	0.350	0.000	0.000	1	no
09_garch_mild	GARCH mild	1000	50	4.00e-04	0.015	0.200	0.200	0.050	0.900	1	no
10_garch_strong	GARCH strong	1000	50	4.00e-04	0.015	0.300	0.400	0.100	0.850	1	no
11_multifactor	Multi-factor K=3	1000	50	4.00e-04	0.015	0.200	0.300	0.000	0.000	3	no
12_regime	Regime-switching	2000	50	4.00e-04	0.015	0.150	0.250	0.000	0.000	1	yes
13_high_ar1	High-persistence AR(1)	1000	50	4.00e-04	0.015	0.700	0.300	0.000	0.000	1	no
14_garch_highvol	High-volatility GARCH	1000	50	4.00e-04	0.015	0.200	0.300	0.150	0.800	1	no

Table 22. Coverage of the date-level Sharpe target by alternative interval procedures.

DGP	blocked	hac_delta	iid	row_naive	stationary
01_iid	0.947	0.952	0.951	0.000	0.947
02_ser_mild	0.938	0.943	0.898	0.000	0.936
03_ser_strong	0.921	0.934	0.734	0.000	0.919
04_xs_mild	0.950	0.950	0.952	0.073	0.950
05_xs_strong	0.934	0.938	0.942	0.262	0.938
06_both_mod	0.937	0.939	0.898	0.088	0.936
07_both_strong	0.934	0.943	0.771	0.178	0.935
08_realistic	0.937	0.941	0.901	0.107	0.937
09_garch_mild	0.926	0.940	0.877	0.110	0.930
10_garch_strong	0.926	0.948	0.862	0.199	0.927
11_multifactor	0.937	0.947	0.895	0.225	0.940
12_regime	0.929	0.934	0.890	0.306	0.929
13_high_ar1	0.899	0.933	0.562	0.137	0.907
14_garch_highvol	0.940	0.948	0.895	0.193	0.941

Note: Row-naive intervals are evaluated against the economically relevant date-level Sharpe target, not against their own row-distribution Sharpe target. Low row-naive coverage therefore records target mismatch and false precision at the portfolio boundary.

Table 23. Row-naive interval behavior when evaluated against the date-level Sharpe target.

DGP	Name	row coverage	bias vs date SR	CI width	mean SE	n
01_iid	IID baseline	0.000	-2.567	0.278	0.071	1000
02_ser_mild	Serial mild	0.000	-2.567	0.278	0.071	1000
03_ser_strong	Serial strong	0.000	-2.568	0.278	0.071	1000
04_xs_mild	Cross mild	0.073	-0.474	0.278	0.071	1000
05_xs_strong	Cross strong	0.262	-0.153	0.278	0.071	1000
06_both_mod	Both moderate	0.088	-0.503	0.278	0.071	1000
07_both_strong	Both strong	0.178	-0.145	0.278	0.071	1000
08_realistic	Realistic	0.107	-0.277	0.124	0.032	1000
09_garch_mild	GARCH mild	0.110	-0.473	0.278	0.071	1000
10_garch_strong	GARCH strong	0.199	-0.268	0.278	0.071	1000
11_multifactor	Multi-factor K=3	0.225	-0.327	0.278	0.071	1000
12_regime	Regime-switching	0.306	-0.092	0.197	0.050	1000
13_high_ar1	High-persistence AR(1)	0.137	-0.338	0.278	0.071	1000
14_garch_highvol	High-volatility GARCH	0.193	-0.325	0.278	0.071	1000

Note: These are not ordinary row-estimator coverage rates; they evaluate the row-naive interval against the economically relevant date-level target.

Table 24. Sampling-boundary stress tests under a null design with no true edge.

Panel	Setting	Row rej.	HAC rej.	Row mean SE	HAC mean SE
A. Same-date correlation (N=50)	rho=0.000	0.097	0.056	0.071	0.598
A. Same-date correlation (N=50)	rho=0.100	0.310	0.064	0.071	0.598
A. Same-date correlation (N=50)	rho=0.300	0.354	0.044	0.071	0.599
A. Same-date correlation (N=50)	rho=0.350	0.368	0.071	0.071	0.597
A. Same-date correlation (N=50)	rho=0.500	0.409	0.047	0.071	0.597
A. Same-date correlation (N=50)	rho=0.700	0.408	0.060	0.071	0.599
B. Selected count (rho=0.35)	N=5	0.192	0.058	0.225	0.598
B. Selected count (rho=0.35)	N=25	0.308	0.052	0.100	0.600
B. Selected count (rho=0.35)	N=50	0.368	0.071	0.071	0.597
B. Selected count (rho=0.35)	N=100	0.421	0.056	0.050	0.598

Note: Row-naive overrejection rises with same-date dependence and selected-count pressure, whereas date-level HAC remains close to nominal size.

Table 25. Monte Carlo rejection rates under null designs with no true edge.

Null design	Row-naive	Date-IID bootstrap	HAC-delta	Moving-block bootstrap	Stationary bootstrap	Romano-Wolf
Dependent null	0.352	0.094	0.066	0.070	0.067	0.068
GARCH null	0.351	0.098	0.063	0.070	0.068	0.067
IID null	0.062	0.060	0.060	0.061	0.062	0.060

Note: Each entry is the fraction of 1000 Monte Carlo replications rejected at nominal 5 percent size. The main null designs use T=1000 dates and N=50 entities. Row-naive treats selected asset-date rows as independent; the other procedures operate on the date-level portfolio return series.

Table 26. Rejection rates across true annualized Sharpe values.

index	true_sharpe	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
0.000	0.000	0.081	0.053	0.058	0.055	0.371	0.056
1.000	0.094	0.105	0.071	0.072	0.072	0.424	0.077
2.000	0.189	0.139	0.088	0.094	0.095	0.495	0.092
3.000	0.378	0.219	0.151	0.160	0.162	0.610	0.162
4.000	0.755	0.459	0.354	0.373	0.376	0.797	0.380
5.000	1.511	0.858	0.788	0.805	0.800	0.985	0.798
6.000	2.833	1.000	1.000	1.000	1.000	1.000	1.000

```
configuration code/campaign_registry.json:  
3a616f119f3fe2603b0663e3f68c499496c37a1781f3003e82814c9269a16259  
generated campaign metadata campaign_metadata.json:  
5cb21ac2cb87cb74a26090a66b28dac6890f76dd1d4bd4391ea6290267810dff
```

Stationarity diagnostics, holdout splits, HAC bandwidth sweeps, prewhitened HAC, fixed-fraction HAC, moving-block bootstrap, and stationary bootstrap results are secondary robustness checks. They are used to identify fragile claims, not to replace the main date-return decision rule.